

Starvation of the bacteria *Vibrio atlanticus* promotes lightning group-attacks on the dinoflagellate *Alexandrium pacificum*

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eLife Assessment

This **important** study **convincingly** shows that *Vibrio* bacteria act as predators of ecologically significant algae that contribute to harmful blooms in the lab, as well as in their natural habitat. While the data strongly suggest that starvation may induce predation, further work is needed to fully establish this link. Similarly, the evidence for a social component in the predation process remains **incomplete**. This study will be very impactful to those interested in the diversity of microbial predator-prey interactions and controlling toxic algal bloom, but the paper could be strengthened by more clearly showing the degree of replication, by better defining the terms used to describe the observed behaviour, and by providing better support for starvation and collective behaviour.

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Abstract

Algae serve as a source of nutrients for bacteria in the marine environment. The interactions between algae and bacteria at the phycosphere interface are known to include mutualism, commensalism, competition or antagonism. Here, based on *in situ* observation and on an *in vitro* interaction study, we report on a novel form of starvation-induced hunting that the cells of selected *Vibrio* species exert on the dinoflagellate species. Results showed that *Vibrio atlanticus* was able to coordinate lightning group attacks then kill the dinoflagellate *Alexandrium pacificum* ACT03. Briefly, the observed coordinated mechanism of algal-killing consists of first, the 'immobilization stage' involving the secretion of algicidal metabolites that disrupt the flagella of the prey. The 'attack stage' resembles the 'wolf-pack attack' strategy,

during which *Vibrios* surrounds algal cells at high density for a brief period without invading them. Finally, the ‘killing stage’ induces the lysis and degradation of the dinoflagellates. By using a combination of biochemical, proteomic, genetic and fluorescence microscopy approaches, we showed that this relationship is not related to the decomposition of algal organic matter, *Vibrio* quorum sensing pathways, to the toxicity of the algae or to the pathogenicity of the bacterium but is conditioned by nutrient stress, iron availability and link to the iron-vibrio-ferritin transport system of *V. atlanticus*. This is the first evidence of a new mechanism that could be involved in regulating *Alexandrium* spp. blooms and giving *Vibrio* a competitive advantage in obtaining nutrients from the environment.

Introduction

Harmful algal blooms (HABs) have experienced an increase in their occurrence, intensity, and geographical distribution on a global scale, resulting in adverse environmental, health, and socioeconomic impacts (Marampouti et al., 2021 [↗](#)). HABs have a considerable impact on human health as a result of direct exposure to volatile toxins or by toxic seafood consumption (Burkholder et al., 2018 [↗](#)). From an ecological point of view, the expansion of HABs can result in the erosion of biodiversity, because they cause massive mortality of marine species and they are generally monospecific in nature (Chai et al., 2020 [↗](#)). In coastal areas, understanding the biological interactions that control toxic algal blooms is therefore a major ecological challenge. Among HABs, a number of *Alexandrium* species have been placed on the list of invasive Mediterranean species. Among them, *Alexandrium pacificum* is a flagellated eukaryotic unicellular organism that form with *Alexandrium tamarense* and *Alexandrium fundyense* the “*Alexandrium tamarense*” complex (Hadjadji et al., 2020 [↗](#)). Since 1998, *A. pacificum* (former *A. catenella*) was monitored by Rephy survey in the Thau lagoon (French Mediterranean) because it produces paralytic shellfish toxins (PSTs) resulting of paralytic shellfish poisoning (PSP) syndrome.

Laania (Laanaia et al., 2013 [↗](#)) showed that in Thau lagoon, a water temperature around 20°C for several days and organic and inorganic nutrients in sufficient concentrations are parameters favoring the developemnt of *A. pacificum*, whose massive blooms occur in autumn.

Algal blooms are seasonal events resulting in a rapid increase in the concentration of a species of algae in an aquatic environment. Depending on the phytoplankton species, the tolerance range of physicochemical parameters is different and influences the time of appearance of blooms (Leblad et al., 2020 [↗](#)).

Interestingly, although the collapse of phytoplankton blooms has been previously attributed to viruses (Pal et al., 2020 [↗](#)), some ecological studies have suggested an important role of algicidal bacteria (Su et al., 2007 [↗](#); Wang et al., 2010 [↗](#)), such as some *Vibrio* spp. (Li et al., 2014 [↗](#); Wang et al., 2020 [↗](#)).

Vibrio (class γ -proteobacteria) are common microorganisms in marine systems worldwide (Baker-Austin et al., 2017 [↗](#); Mavian et al., 2020 [↗](#)), where they are important components of the food chain, particularly in biodegradation, nutrient regeneration and biogeochemical cycles (Oberbeckmann et al., 2012 [↗](#)). *Vibrio* is one of the most studied bacterial taxa due to their ubiquity in coastal marine systems and their capacity to cause infections in humans and animals, leading sometimes to epizootic or zoonotic epidemics (LeRoux et al., 2015 [↗](#); Mavian et al., 2020 [↗](#)). *Vibrio* are extremely adaptable to their environment (Johnson, 2013 [↗](#)). The main factors influencing their occurrence and distribution in water are temperature, salinity, nutrient availability (Wang et al., 2020 [↗](#)), multiple strategies such as biofilm formation on biotic and abiotic surfaces (Espinoza-Vergara et al., 2020 [↗](#)), or interactions with a multitude of other organisms such as eukaryotic predators (Drebes Dörr and Blokesch, 2020 [↗](#)) or plankton (Lopez-Joven et al., 2018 [↗](#)) are used by *Vibrio* in the environment. There is also evidence that global

climate change has increased *Vibrio*-associated illnesses affecting humans and animals (Brumfield et al., 2021 [↗](#); Muhling et al., 2017 [↗](#)). However, the drivers and dynamics of *Vibrio* survival and propagation in the marine environment are not yet fully understood.

A substantial number of research articles have highlighted the potential of γ -proteobacteria to exert algicidal activity against dinoflagellates, supporting the hypothesis that γ -proteobacteria such as *Vibrio* play a role in the control of algae blooms *in situ* (Coyne et al., 2022 [↗](#)). However, the mechanisms behind *Vibrio*-driven algal lysis in the environment remain to be elucidated. Particularly, it is unclear how in the water column, algicidal compounds secreted by bacteria can concentrate around the algae to exert their lytic effect.

Based on observations from the natural environment showing a potent relationship between *Vibrio* and *Alexandrium* algae bloom events, this study aims to determine *in vitro*, the main factors implicated in this relationship. Using a combination of biochemical, proteomic, genetic, and fluorescence microscopy approaches, we explored the role of algal toxicity, bacterial pathogenicity and the quorum sensing pathway on this relationship and showed the important role of nutrient stress and the iron uptake pathway in this unique *Vibrio*/*Alexandrium* predator/prey interaction.

Methods

Quantification of *Alexandrium* algae and *Vibrio* bacteria in the environment

Seawater samples were collected in the Thau Lagoon (southern France, a shallow Mediterranean ecosystem open to the sea (Abadie et al., n.d.), during spring and autumn 2015. Briefly, samples were collected from the subsurface (-50 cm) near an oyster table at a phytoplankton surveillance site (part of the REPHY network, N 43°26.058' and E 003°39.878'). Once a week during spring and autumn 2015, during field sampling campaigns, 20 L of water was filtered on board through a 180 μ m pore-size nylon membrane. At the laboratory, according to Lopez-Joven et al. (Lopez-Joven et al., 2018 [↗](#)) seawater was fractionated into two size classes as follows: 2 L of the above filtrate was filtered through a 0.8 μ m pore-size polycarbonate Whatman Nuclepore membrane to obtain organisms in the 0.8–180 μ m range corresponding to plankton-associated *Vibrio* and living *Alexandrium* forms. Then, the filtrate from the 0.8 μ m membrane was filtered again through a 0.2 μ m pore-size polycarbonate Whatman Nuclepore membrane until the membrane was saturated. *Alexandrium* cells, ranging from 25 to 40 μ m, belong to the microphytoplankton and are therefore retained in the 0.8–180 μ m fraction. Any *Vibrio* cells potentially associated with or attached to *Alexandrium* cells will also be retained in this fraction. *Vibrio* cells are approximately 0.5–0.8 μ m-thick. The fraction between 0.2 and 0.8 μ m therefore includes the free-living *Vibrio*. The bacterial population collected on 0.8- μ m-pore-size filters was designated the particle-associated community, and the population on 0.2- μ m-pore-size filters was designated as the free-living community. Membranes (in triplicate) were then conserved in 500 μ L of 100% EtOH at -20°C. Environmental DNA (eDNA) was extracted from the MF Millipore membrane using the Macherey-Nagel NucleoSpin Tissue Kit and resuspended in 100 μ L of water. The samples were then stored at -20°C after eDNA quantity and purity were assessed using a NanoDrop system (NanoDrop Technologies, Wilmington, DE, USA). PCR amplification reactions were done on a Roche LightCycler 480 Real-Time thermocycler (qPHD platform, University of Montpellier, France) using specific primer pairs (Table 3 [↗](#)). Typically, the reactions contained 1 μ L of template DNA (the DNA concentration for all samples varied from 1 to 40 μ g mL⁻¹), 0.5 μ L of each primer (3.33 μ M) and 4 μ L of reaction mixture (SYBR Green Master Mix) in a total volume of 6 μ L. The reaction parameters were as follows: 5 minutes at 95°C (initial denaturation) and 40 cycles of 10 s at 95°C (denaturation), 10 s at the corresponding hybridization temperature (Table 3 [↗](#)) and 10 s at 72°C (elongation). Melting curve profiles were generated by increasing the temperature from 65°C to 95°C at 0.5°C s⁻¹. Amplification products were analysed using LightCycler software (Roche Diagnostics). *Vibrio* spp. and *A.*

pacificum and *A. tamarensis* were quantified by constructing calibration curves based on DNA from the *V. atlanticus* LGP32 reference strain (former *V. tasmaniensis* LGP32) and from the *A. pacificum* reference strain (ACT03: *A. catenella* strain isolated from Thau in 2003) and the *A. tamarensis* reference strain (ATT07: *A. tamarensis* isolated from Thau in 2007) (not shown).

Strains and growth conditions

Vibrio strains. Wild-type and isogenic mutants of *Vibrio atlanticus* LGP32 (Table 1) were used in this study. Deletion-mutants included $\Delta luxR$, $\Delta luxP$, $\Delta luxM$ and $\Delta pvuB$ isogenic strains. The $\Delta pvuB$ mutant was constructed here by allelic exchange as described previously by Le Roux (Le Roux et al., 2007). We also used *V. atlanticus* LGP32 carrying the pSW3654T-GFP plasmid (Le Roux et al., 2007), hereafter referred to as *V. atlanticus* LGP32-GFP. Bacterial strains were grown at $22 \pm 1^\circ\text{C}$ in Zobell medium (0, 38 μM iron (III)). When needed, 25 $\mu\text{g mL}^{-1}$ chloramphenicol (Cm) was added to cultures of *V. atlanticus* LGP32-GFP (Le Roux et al., 2007). Oligonucleotides used for RNA expression analysis are specified in table 3.

Phytoplankton strains. Non-axenic phytoplankton species (Table 2) were grown in batch culture in enriched natural sea water (ENSW, 6,55 μM iron (III)) with a salinity of 36 practical salinity units (PSU) at $22 \pm 1^\circ\text{C}$ under cool white fluorescent illumination (100 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$) and a 12 h:12 h light:dark cycle (Harrison et al., 1980). The algae were used for experiments in their exponential growth phase.

Co-culture assay

For each tested phytoplankton species, 2.10^4 cells harvested in their exponential growth phase were placed in 20 mL of ENSW medium in a 50 mL suspension culture flask (Cellstar® PS, Greiner bio-one). After incubation for 24 h at $22 \pm 1^\circ\text{C}$ under cool white fluorescent illumination, 40 μL of *Vibrio* strains grown for 12, 36, 60 and 156 h in Zobell medium or in Zobell medium supplemented with FeCl_3 (6 μM iron (III)) or with H_3BO_3 (0.47 mM) or the corresponding culture supernatant were added to the algal culture. After incubation at $22 \pm 1^\circ\text{C}$ under cool white fluorescent lights, living, non-swimming, group-attacked and lysed phytoplankton cells were counted in a sedimentation chamber under an inverted microscope. The number of lysed cells corresponded to algae showing disrupted membranes. Non-swimming algae were not counted as lysed cells. For *Vibrio* analysis, 100 μL of a 1:10 serial dilution mixture in ENSW (from 10^{-2} to 10^{-10}) was plated on *Vibrio* Selective TCBS (thiosulfate-citrate-bile salts-sucrose) agar (in triplicate). After incubation for 24 h at $22 \pm 1^\circ\text{C}$, the number of living *Vibrio* cells was determined by counting colony-forming units (CFUs). Data are the means of three independent experiments.

Microscope observations

The dynamic of the interaction between *A. pacificum* ACT03 and *V. atlanticus* LGP32, which is an oyster pathogen isolated from the French Atlantic coast (Gay et al., 2004) and present in the Thau Lagoon, south of France (Lopez-Joven et al., 2018) was surveyed. As the algal strain used in the study is not axenic, means that additional bacteria, other than the *V. atlanticus* LGP32, are present in the experiments. To elucidate the interaction without thoroughly accounting for the non-axenic cells, interaction was observed under a Zeiss Axio upright fluorescence microscope equipped with an AxioCamMRm 2 digital microscope camera using *V. atlanticus* LGP32 tagged with green fluorescent protein (GFP). Lasers were used at excitation wavelength (λ_{ex}) 488 nm for GFP (emission wavelengths (λ_{em}): 505–530 nm) and λ_{ex} 532 nm for plankton chloroplasts (λ_{em} 560–630 nm). Images were taken sequentially to avoid cross-contamination between fluorochromes. Sequences of images were merged during the *Vibrio*-*Alexandrium* interaction using ZEN 2012 (blue edition) software. Interaction events between *Vibrio* strains and phytoplankton strains were also observed under a Leica TCS SPE confocal laser scanning system connected to a Leica DM 2500 upright microscope camera (Montpellier RIO Imaging Platform, University of Montpellier, France).

Vibrio species	Strains	Virulence for fish or invertebrates	References	Secretion of algicidal compounds (This study)	Group attack (This study)
<i>Vibrio atlanticus</i> LGP32	WT	+	(Gay et al., 2004)	+	+
//	WT + pSW3654T-GFP	+	(Le Roux et al., 2007)	+	+
//	ΔLuxM	ND	Ifremer Institute, France	+	+
//	ΔLuxS	ND	//	+	+
//	ΔLuxR	ND	//	+	+
//	ΔPvuB	ND	This work	+	-
<i>Vibrio tasmaniensis</i>	J5-9	+	(Lemire et al., 2015)	+	+
//	LMG20012 ^T	-	(Thompson et al., 2003)	+	+
<i>Vibrio crassostreae</i>	J2.9	+	(Lemire et al., 2015)	+	+
//	J2-8	-	//	+	+
<i>Vibrio fischeri</i>	ES114	ND	(Mandel et al., 2008)	+	+
<i>Vibrio harveyi</i>	ATCC14126	+	(Liu et al., 1996)	+	+
<i>Vibrio aestuarianus</i>	janv-32	+	(Labreuche et al., 2010)	+	+

Table 1.

Ability of *Vibrio* spp. and strains to secrete algicidal compounds and attack *Alexandrium pacificum* LGP32 in groups.

NT: not determined.

Dinoflagellates species	Strains	Toxicity for human	References	Degraded flagella (This study)	Attacked in groups (This study)	Lysed cells (This study)
<i>Alexandrium pacificum</i>	ACT03, Thau, France	+	(Laabir et al., 2011)	+	+	+
<i>Alexandrium catenella</i>	Bizerte, Tunisia	+	(Fertouna-Bellakhal et al., 2015)	+	+	+
//	F3-9F, Tarragona, Spain	ND	//	+	+	+
//	C10-5, Annaba, Algeria	+	(Hadjadji et al., 2020)	+	+	+
<i>Alexandrium tamarense</i>	ATT07, Thau, France	-	(Rolland et al., 2012)	+	+	+
<i>Alexandrium</i> spp.	Golf of Tunis, Tunisia	ND	Algal collection university of Montpellier, France	+	+	+
//	Bizerte, Tunisia	ND	//	+	+	+
//	Mediterranean coast, Morocco	ND	//	+	+	+
<i>Prorocentrum lima</i>	PLBZT14, Bizerte, Tunisia	+	(Ben-Gharbia et al., 2016)	-	-	-
<i>Coolia monotis</i>	CMBZT14, Bizerte, Tunisia	ND	//	-	-	-
<i>Vulcanodinium rugosum</i>	IFR-VRU-01, Ingril, France	+	(Abadie et al., 2015)	-	-	-
<i>Karenia selliformis</i>	Golf of Gabes, Tunisia	NT	Algal collection university of Montpellier, France	-	-	-
<i>Scirpsiella trochoidea</i>	Mellah Lagoon, Algeria	-	//	-	-	-
<i>Gyrodinium impudicum</i>	Golf of Tunis, Tunisia	-	//	-	-	-
<i>Amphidium carterae</i>	SAMS, Scotland	NT	SAMS laboratory, Scotland	-	-	-
<i>Gymnodinium catenatum</i>	M'diq Bay, Morocco	+	(Leblad et al., 2020)	+	+	+

Table 2.

Ability of *Vibrio Atlanticus* LGP32 to degrade flagella, attack in group and lysed the targeted dinoflagellates spp.

commonly found in the Mediterranean Sea. ND not determined.

Comparative proteomic analysis

Vibrio sampling and protein extraction. *V. atlanticus* LGP32 was grown for 60 h at 22°C in artificial seawater (high nutrient stress, cond. 1) or 12 h at 22°C in Zobell media (low nutrient stress, cond. 2). After 10 min centrifugation at 8000 rpm, crude protein extracts of *V. atlanticus* LGP32 in each culture condition (triplicates) were obtained by sonication on ice at 20% amplitude for 20 s in 200 µL of ice-cold denaturing buffer (7 M urea, 2 M thiourea, 4% CHAPS in 30 mM Tris-HCl, pH 8.5) and clarified by centrifugation at 2000 x g, 15 min, 4°C. The protein concentration of the supernatant was estimated using the 2D Quant Kit (Cytiva™, MERCK) and samples were stored at -80°C until use.

Bi-dimensional gel electrophoresis (2D gel). Proteins extracts were individually analysed on 2D gel electrophoresis (6 gels per condition each corresponding to different biological replicates). To do so, 100 µg of proteins from each extract was added to rehydration buffer (7 M urea, 2 M thiourea, 4% CHAPS, 65 mM DTT) for a total volume of 350 µL. They were then individually loaded onto 17 cm isoelectric focusing strips (Bio-Rad) with a stabilized non-linear pH ranging from 3 to 10. Due to the high complexity of the protein profile in the acidic part (left) of the gel pH 3–10 (**Fig. S3** [↗](#)), we conducted additional ‘close-up’ analyses in gels using 17 cm isoelectric focusing strips (Bio-Rad) with a narrower, stabilized pH gradient ranging from 4 to 7. Strips were rehydrated passively for 5 h at 22°C, followed by active rehydration for 14 h under a 50 V current at 22°C (to help large proteins enter the strips). Thereafter, isoelectrofocusing was carried out using the following programme: 50 V for 1 h, 250 V for 1 h, 8000 V for 1 h and a final step at 8000 V for a total of 140 000 V.h with a slow ramping voltage (quadratic increasing voltage) at each step. Focused proteins were reduced by incubating the strip twice in equilibration buffer (1.5 M Tris, 6 M urea, 2% SDS, 30% glycerol; bromophenol blue, pH 8.8) containing DTT (130 mM) at 55°C. Then, they were alkylated by incubation with equilibration buffer containing iodoacetamide (135 mM) on a rocking agitator (400 rpm) at room temperature protected from light. Proteins were also separated according to their molecular weight (second dimension) on 12% acrylamide/0.32% piperazine diacrylamide gels run at 25 mA per gel for 30 min followed by 75 mA per gel for 8 h using a Protean II XL system (Bio-Rad). Gels were stained using an MS-compatible silver staining protocol and scanned using a ChemiDoc MP Imaging System (Bio-Rad) associated with Image Lab software version 4.0.1 (Bio-Rad).

Comparative bioinformatics analysis of 2D gels. Twelve gels (six per condition) were selected for comparative analysis on PD-Quest v. 7.4.0 (Bio-Rad) to identify changes in protein abundance between the proteomic profiles of *V. atlanticus* LGP32 cultured in contrasting nutrient conditions (ENSW/Zobell). Spots whose mean intensity across six replicates per strain was two times higher or lower than those from the other strain, with a $P < 0.01$ (Mann-Whitney U-test), were considered significantly different in terms of abundance between the two conditions (quantitative difference). Differentially represented spots were then excised from the gels, destained, trypsin-digested and the obtained peptides were identified by tandem mass spectrometry (MS-MS) using the PISSARO platform facility (University of Rouen, France). To identify protein(s) present in each spot, the obtained peptides were compared with *V. atlanticus* LGP32 reference genome (<https://vibrio.biocyc.org/> [↗](#)). The genes whose peptides matched strongly were retrieved and used for an BLASTx query against non-redundant databases to determine the protein identity of the best match. A gene was considered as strongly matched when at least two peptides matched the sequence with a coverage of > 6%. Their theoretical isoelectric point (pI) and molecular weights were also calculated using the ExPASy server (<https://www.expasy.org/> [↗](#)) to compare them with the location of the spot on the gel. Altogether, these complementary analyses made it possible to characterize the protein identity of each spot with confidence.

Gene expression analysis

Vibrio sampling and RNA extraction. *V. atlanticus* LGP32 was grown in Zobell media for 36 h and 60 h at 22°C (decline phase of growth, nutrient stress) or for 12 h at 22°C (exponential growth phase, poor nutrient stress). Total RNA was isolated from *V. atlanticus* LGP32 using the standard TRIzol method (Invitrogen Life Technologies SAS, Saint-Aubin, France) and then treated with DNase (Invitrogen) to eliminate genomic DNA contamination. After sodium acetate precipitation, the quantity and quality of the total RNA were determined using a NanoDrop spectrophotometer and agarose gel electrophoresis. Following heat denaturing (70°C for 5 minutes), reverse transcription was performed using 1 µg of RNA prepared with 50 ng µL⁻¹ oligo-(dT) 12–18 mer in a 20-µL reaction containing 1 mM dNTPs, 1 unit µL⁻¹ RNaseOUT and 200 units µL⁻¹ Moloney murine leukaemia virus reverse transcriptase (M-MLV RT) in reverse transcriptase buffer, according to the manufacturer's instructions (Invitrogen Life Technologies SAS, Saint-Aubin, France).

PCR amplification. Amplification reactions were analysed using a Roche LightCycler 480 Real-Time thermocycler (Bio-Environnement platform, University of Perpignan, France). In this study, several PCR primer pairs were designed using Primer3 software (optimal primer size: 20 bases; T_m: 60°C; primer GC%: 50; 2GC clamp and product size range: 150–200 bp) and calibrated with *V. atlanticus* LGP32 genomic DNA (Table 1). To determine the qPCR efficiency of each primer pair used, standard curves were generated using seven serial dilutions of genomic DNA (10⁻², 10⁻³, 10⁻⁴, 10⁻⁵, 10⁻⁶, 10⁻⁷ and 10⁻⁸) (not shown); the qPCR efficiencies of the tested genes varied between 1.85 and 2.08 (Table 3). For gene expression, reverse transcription was performed with 1 µg of total RNA using random hexamers and SuperScript IV reverse transcriptase (Invitrogen). The total qPCR reaction volume was 10 µL and consisted of 5 µL of cDNA (diluted 1:5), 2.5 µL of SensiFAST SYBR No-ROX Mix (Bioline) and 100 nM or 300 nM PCR primer pair (Table 3). The reaction parameters were as follows: 2 min at 95°C (initial denaturation) and 40 cycles of 5 s at 95°C (denaturation), 10 s at 59°C (annealing) and 20 s at 72°C (elongation). The specificity of each PCR was checked by measuring fluorescent signals during melting curve analysis (PCR product heated from 65°C to 95°C continuously and slowly at 0.1°C s⁻¹). Relative expression was calculated by normalization to the expression of two constitutively expressed housekeeping genes, namely, 6PKF (VS_2913) and CcmC (VS_0852), using the delta-delta threshold cycle (ΔΔCt) method (Pfaffl, 2001).

Detection of quorum-sensing signalling molecules

Vibrio culture. To detect the QS molecules (AI-2, AI-1 and CAI-1) *V. atlanticus* LGP32 was grown in Zobell media for 12 h (exponential growth phase, control) and 60 h (decline phase of growth, nutrient stress).

AI-2 analysis. Bioluminescence assay using the QS bioluminescent of *Vibrio campbellii* MM32 (*luxN::Cm*, *luxS::Tn5Kan*) was used to detect AI-2 molecules in culture supernatants. Briefly, *Vibrio* cultures were centrifuged at 17,000 × *g* for 10 min, and the resulting supernatants were filtered on 0.22 µm. Then 20 µL of the filtrates were mixed with 180 µL of *V. campbellii* MM32 diluted 1:5000 then incubated at 30°C and 100 rpm. Luminescence and cell density (OD₆₂₀) were collected in triplicate and analysed according to Tourneroche et al. (Tourneroche et al., 2019).

AI-1 and CAI-1 extraction and LC-MS analysis. Chemical analyses were conducted with a Q Exactive Focus Orbitrap System coupled to an Ultimate 3000™ ultrahigh-performance liquid chromatography (UHPLC) system (Thermo Fisher Scientific) according to Rodrigues et al. (Rodrigues et al., 2022). Briefly, ethyl acetate (2 mL) was added into each culture (2 mL). This mixture was shaken overnight at room temperature (150 rpm). The two phases were then separated and the aqueous phase was extracted once again. The two obtained organic phases were pooled and the solvent was evaporated under vacuum. The crude extracts were dissolved in 500 µL LC-MS grade methanol for analysis. All experiments were conducted in triplicate. Analyses of

Species	genes	Primers Sequences	Tm (°C)	Efficiency	References
<i>Alexandrium pacificum</i> (ACT03)	18S – 28S rRNA ITS region	TGATATTGTGGGCAACTGTAA AACATCTGTTAGCTCACGGAA	54		(Genovesi et al., 2011)
<i>Alexandrium tamarense</i> (ATT07)	18S – 28S rRNA ITS region	TGGTAATTCTTCATTGATTACAATG AACATCTGTTAGCTCACGGAA	54		//
<i>Vibrios spp.</i>	16S	CGGTGAAATGCGTAGAGAT TTACTAGCGATTCCGAGTTC	62		(Kitatsukamoto et al., 1993)
<i>Vibrio atlanticus</i> LGP32	LuxN (VS_II0260)	CACTTGCTAGTATCATCGC ATCGAGTTAGCAAGAGCAC	60	1,92	This work
//	LuxM (VS_II0261)	TCCACTTATCACAAACAGG ACTGTACTTCCATTTGTCG	60	1,91	//
//	LuxP (VS_II0355)	AAGTTCAGGATGAACCTATC CAAAGAGATACTTTGCTGAG	60	1,89	//
//	LuxS (VS_2562)	ACTCTCGAGCACCTATACG GAAGGCGTACCAATCAAGC	60	1,85	//
//	CqsS (VS_1725)	GACATCTATTGATGTTATGC TCACCCACTTCACGTAACCTG	60	1,91	//
//	PvsA (VS_II0355), Vibrio ferrin biosynthesis protein	CAGAGCAAGAGCTAGAACC TCGTTGAGAACCTGACGAG	59	1,91	//
//	PvuB (VS_II1126), ABC transporter vibrio ferrin uptake FecB	TAGTGCAACCATGGGAATCG TAAACCGTACGTAGACGCTC	57	2,01	//
//	PvuA2 (VS_II1127), Vibrio ferrin receptor FecA	GGAGCTACAAGCATTGCTTC TTCGTCATATGGTCGCTTCG	57	2,08	//
//	Housekeeping gene 1, CcmC (VS_0852)	ATTGCCGCCTTTATCGGTTT CAAGCACCCACATTGGTTT	60		(Vanhove et al., 2016)
//	Housekeeping gene 2, 6-phosphofructokinase (VS_2913)	GCCGTCACCTGTGGTGACCTT TGCTTCTGCCTTTCGCAAT	60		//

Table 3.

Oligonucleotide sequences of primers used for RNA expression analysis.

extracts and standards (3 μL injected) were performed in electrospray positive ionization mode in the 50–750 m/z range in centroid mode. The parameters were as follows: spray voltage: 3 kV; sheath flow rate: 75; aux gas pressure: 20; capillary temperature: 350°C; heater temperature: 430°C. The analysis was conducted in Full MS data-dependent MS2 mode (Discovery mode). Resolution was set to 70,000 in Full MS mode, and the AGC (automatic gain control) target was set to 1×10^6 . In MS2, resolution was 17,500, AGC target was set to 2×10^5 , isolation window was 0.4 m/z , and normalized collision energy was stepped to 15, 30 and 40 eV. The UHPLC column was a Phenomenex Luna Omega Polar C18 1.6 μm , 150 x 2.1 mm. The column temperature was set to 42°C, and the flow rate was 0.4 mL min^{-1} . The solvent system was a mixture of water (A) with increasing proportions of acetonitrile (B), with both solvents modified with 0.1% formic acid. The gradient was as follows: 1% B 3 min before injection, then from 1 to 15 min, a gradient increase of B up to 100% (curve 5), followed by 100% B for 5 min. The flow was injected into the mass spectrometer starting immediately after injection. All data were acquired and processed using FreeStyle 1.5 software (Thermo Fisher Scientific).

Chemicals and solvents. N-acyl-homoserine lactones (AHL) were obtained from Cayman Chemical (Ann Arbor, MI, USA). Stock solutions were obtained by dissolving standards in methanol or dichloromethane (C18-AHL) at a concentration of 1 mg mL^{-1} and stored at -80°C. Standard solutions for UHPLC-high-resolution tandem mass spectrometry (HRMS) analyses were prepared by diluting each individual standard solution with methanol in order to obtain a concentration range from 2000 to 20 ng mL^{-1} . LC-MS grade methanol, acetonitrile and formic acid were purchased from Biosolve (Biosolve Chimie, Dieuze, France), analytical-grade ethyl acetate was obtained from Sigma-Aldrich. Pure water was obtained from Elga Purelab Flex System (Veolia LabWater STI, Antony, France).

Nature of lytic compounds secreted by *V. atlanticus* LGP32

To determine the nature of the lytic compounds secreted, *V. atlanticus* LGP32 grown for 60 h in Zobell media at 22 °C was filtered through a 10 kDa membrane (Amicon® Ultra-4 filter unit). The eluate containing molecules with MW below 10 kDa was then incubated in a water bath at 100°C for 30 min. Boiled filtrates (0.1 mL) were subsequently used to inoculate *A. pacificum* (ACT03 strain) cultures, then lytic activity was observed under the Leica TCS SPE confocal laser scanning system. Zobell media with the same treatment was used as control.

Statistical analysis

Environmental data. Statistical analyses were performed using R 3.6.3 software. The relationship between *Alexandrium* and *Vibrio* was explored separately in spring and autumn. We used a generalized linear model specifying a Gaussian family. For spring, the dataset for salinity and temperature was complete (14 periods of observation). One influential period was detected and removed from the dataset whenever no dead *Alexandrium* cells were observed. The effects of explanatory variables such as $\log_{10}(\text{Vibrio}+1)$, salinity and temperature were centred, reduced and tested as fixed effects with a linear relationship. Model selection was performed using the Akaike information criterion corrected for small sample size (AIC_c). Models were considered different whenever the difference between their AIC_c value and the lowest AIC_c value (ΔAIC_c) was lower than 2 (Burnham and Anderson, 2002). *Alexandrium* distribution and model residuals were checked for normal distribution assumptions (QQ plot and Shapiro-Wilk test). For autumn, the dataset was complete for 10 periods of observation. Salinity and temperature were missing for three periods. We explored the relationship between *Alexandrium* and *Vibrio* alone using the method detailed above for spring.

In vitro data. Statistical analyses were performed using one-way ANOVA (analysed by pair) followed by Tukey's test (Statistica 10.0 software, StatSoft, Maison-Alfort, France). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$

Results

Concomitant occurrence of *A. pacificum* ACT03, *A. tamarense* ATT07 and free-living *Vibrio* spp. in the Thau lagoon

In the spring and autumn of 2015, in the Thau Lagoon ([Fig. 1A](#)), we detected *Alexandrium* algae (*A. pacificum* ACT03 and *A. tamarense* ATT07, both alive and in degraded cell forms) and free-living *Vibrio*, but no plankton-associated *Vibrio* were observed ([Fig. 1B](#), Data S1). Using model selection based on AICc, we found no significant relationship between *Alexandrium* (*A. pacificum* ACT03, *A. tamarense*) and *Vibrio* Spp. abundances in autumn. This result is consistent with the difficulty that *Vibrio* has in growing at temperatures below 20°C and with the complex interacting factors driving bloom dynamics ([Laanaia et al., 2013](#)). Interestingly, in spring 2015, the mean densities of *Alexandrium* and of free-living *Vibrio* were positively correlated. The lowest AICc was obtained with the model explaining degraded form of *Alexandrium* density based on the free *Vibrio* density ([Fig. 1C](#)). Given that, this model is not so different from the model with only the intercept, but better than any other linear combination with other potentially interfering drivers, such as temperature and salinity ([Fig. 1D](#)), we searched for evidence of a relationship between *Vibrio* and *Alexandrium* by studying their interaction *in vitro*.

Vibrio atlanticus LGP32 feeds on *Alexandrium pacificum* ACT03

To investigate whether *Alexandrium* interacts with *Vibrio*, we incubated in Enriched Natural SeaWater (ENSW) *A. pacificum* ACT03 with *V. atlanticus* LGP32 previously grown for 12 hours in Zobell media. In interaction with *V. atlanticus* LGP32, *A. pacificum* ACT03 cell abundance decreased significantly from 2.10×10^4 cells mL⁻¹ to 1.07×10^4 cells mL⁻¹ after 48 h ([Fig. 2A](#)), while the *V. atlanticus* LGP32 concentration grew significantly after 26 h of interaction, reaching a maximum peak density of 7.67×10^7 CFU mL⁻¹ at 34 h ([Fig. 2B](#)). In the control experiment where *A. pacificum* ACT03 was cultured alone in ENSW, the algal concentration remained stable over time ([Fig. 2A](#)) and no bacteria were on the corresponding TCBS plates (*Vibrio* selective medium). In the control where *V. atlanticus* LGP32 was cultured alone in ENSW, the concentration of the bacteria tended to increase during the first 24 h (from 2.0×10^5 to 7.0×10^5 CFU mL⁻¹) and then decreased at 48h, below the initial starting concentration ([Fig. 2B](#)). These results show that the interaction between *V. atlanticus* LGP32 and *A. pacificum* ACT03 leads to a decline in the algal population and promotes the growth of *V. atlanticus* LGP32. This suggests that *V. atlanticus* LGP32 is able to feed on *A. pacificum* ACT03.

V. atlanticus LGP32 performs group-attacks on *A. pacificum* ACT03

Epifluorescence microscopy observation of GFP-labelled *V. atlanticus* LGP32 (previously grown in Zobell medium) in interaction showed that *A. pacificum* ACT03 cells that had lost their motility were attacked individually by groups of *V. atlanticus* LGP32 before being lysed (Movie 1). Attacks were extremely rapid, with empty thecae (algal envelopes) observed in the medium after less than 60 s (Movie 2). During the attack, *V. atlanticus* LGP32 did not invade the algal cell but remained clustered on the cell surface ([Fig. 3A2](#)).

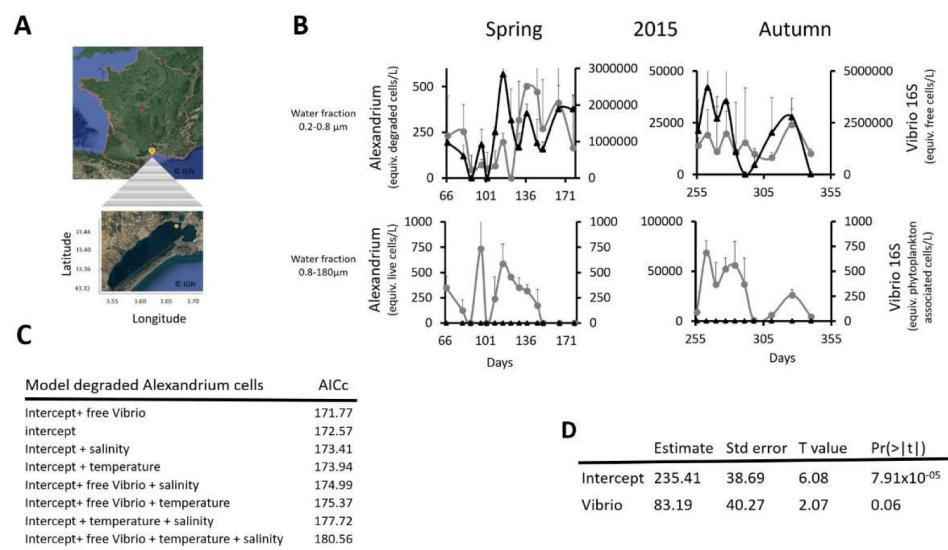


Figure 1.

Dynamics of *Alexandrium* and *Vibrio* in the environment.

(A) Location of the monitoring station in the Thau Lagoon (southern France). **(B)** Mean abundance (DNA equiv.) of *Vibrio* spp. (16S) and *Alexandrium* spp. (*A. pacificum* ACT03 + *A. tamarense* ATT07). *Vibrio* cells (black line with diamond dot) and degraded *Alexandrium* cells (grey line with round dot) were evidence in the 0.2–0.8 μ m fraction (free *Vibrio* fraction) in spring and autumn 2015. Living *Alexandrium* cells (grey line with rounddot) but no plankton-associated *Vibrio* spp. (black line with diamond dot) were evidence in the 0.8–180 μ m in spring and autumn. **(C)** Result of Akaike information criterion (AICc) models tested to explain the mean value of degraded *Alexandrium* cells (dead cells) in spring. **(D)** Wald test of the AICc model attributing the mean value of degraded cells of *Alexandrium* in spring to free *Vibrio*.

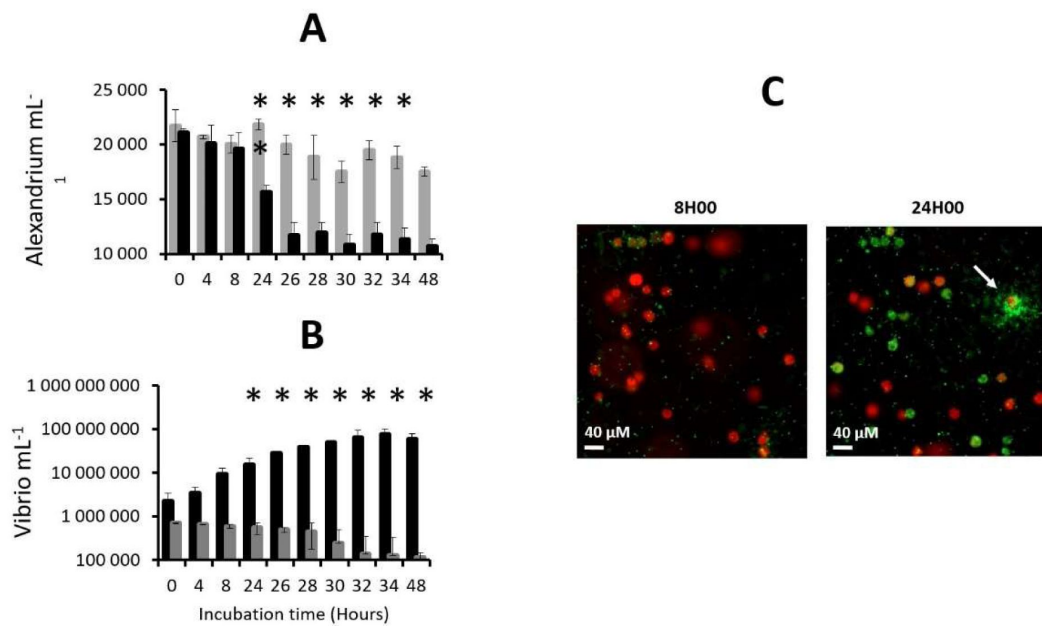


Figure 2.

Incubation of *Vibrio atlanticus* LGP32 and *Alexandrium pacificum* ACT03 in enriched natural seawater (ENSW).

(A) *A. pacificum* ACT03 cultured alone (grey bar) and incubated with *V. atlanticus* LGP32 (black bar) in ENSW. **(B)** *V. atlanticus* LGP32 cultured alone (grey bar) and incubated with *A. pacificum* ACT03 (black bar) in ENSW. **(C)** Images of the interaction of *V. atlanticus* LGP32-GFP with *A. pacificum* ACT03 taken at 8h00 and 28h00 of co-culture. *V. atlanticus* LGP32 (small green cells), living *A. pacificum* ACT03 cells (large red cells) and degraded *A. pacificum* ACT03 cells (large green cells). *P < 0.01 (analysis by pairs).

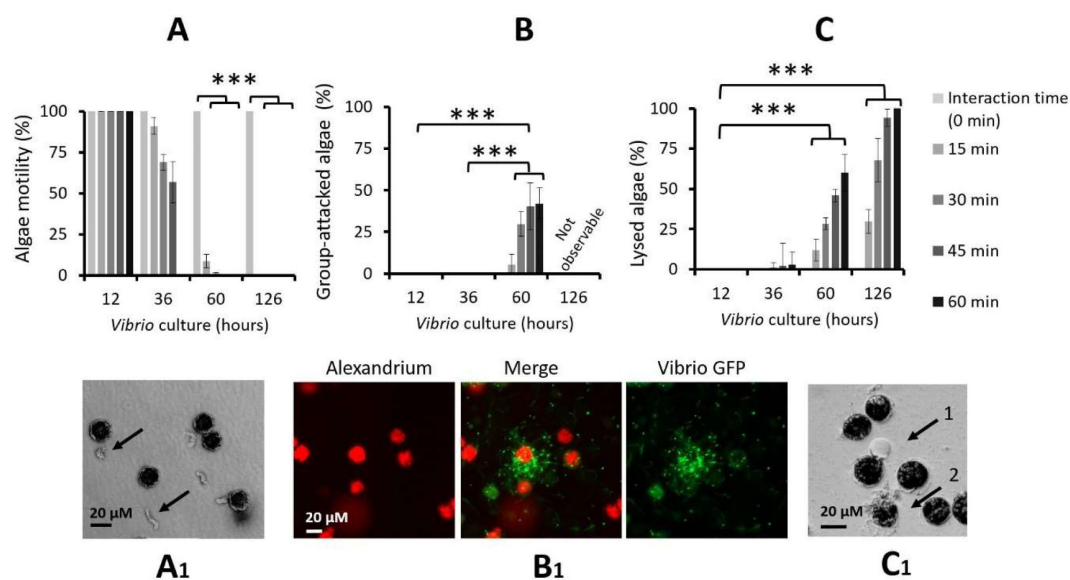


Figure 3.

Role of *Vibrio atlanticus* LGP32 starvation in the interspecific interaction process.

Experiments were conducted by incubating *A. pacificum* ACT03 with *V. atlanticus* LGP32 previously grown for 12, 36, 60 and 126 h in Zobell medium. **(A)** Percentage of motile *A. pacificum* ACT03. **(B)** *A. pacificum* ACT03 group-attacked by *V. atlanticus* LGP32 and **(C)** *A. pacificum* ACT03 lysis after 0, 15, 30, 45 and 60 min of interaction. Corresponding pictures showing **(A1)** Black arrows indicate unhooked and degraded flagellum from *A. pacificum* ACT03 flagellum, **(B1)** Image of the interaction of *V. atlanticus* LGP32-GFP with *A. pacificum* ACT03 taken at 8h00 of co-culture. *V. atlanticus* LGP32 (small green cells), living *A. pacificum* ACT03 (large red cells) and dead *A. pacificum* ACT03 (large green cell). **(C1)** Black arrow 1 indicate vesicle formation on *A. pacificum* ACT03 cell and black arrow 2 indicate exploded *A. pacificum* ACT03 cell. ***P < 0.001.

Group attack of *A. pacificum* ACT03 is activated by *V. atlanticus* LGP32 starvation

In 2002, Martin hypothesized that nutritional stress induces bacteria to lyse algae (Martin, 2002 [↗](#)). To test this hypothesis, we monitored *V. atlanticus* LGP32 behaviour in response to starvation (Fig. 3 [↗](#)). We observed that *V. atlanticus* LGP32 in exponentially growth phase (12 h of culture in Zobell medium) did not interact with *A. pacificum* ACT03 cells for the first hour of contact (Fig. 3A, B, C [↗](#)). In contrast, *V. atlanticus* LGP32 in the decline phase (36 h of culture in Zobell medium) induced a significant decrease in the number of motile algae cells by 8.9% after 15 min and by 43.3% after 60 min (Fig. 3A [↗](#)). This phenomenon corresponded to the degradation and/or disruption of algal flagella (Movie 3). The flagella no longer functioned correctly, which caused erratic swimming of the algae (Video 3, left cell). This was followed by a complete cessation of swimming. When the flagellum detached from the algae (Video 3, right cell), the attack occurred. With starved *V. atlanticus* LGP32 (60 h of culture in Zobell medium), algae immobilization was fast and significant (91.4% in 15 min, Fig. 3A [↗](#)), and algae were attacked individually being targeted by groups of *V. atlanticus* LGP32 (Movie 4). The percentage of cells attacked and killed peaked at 30% after 15–30 min of contact (Fig. 3B [↗](#)) and then decreased. After 1 h, attacks had stopped with approximately 40% of the algal cells still alive (Fig. 3C [↗](#)). Although it remains unclear whether the attacks occur during a specific phase of growth, it is evident that the cells are already weakened before attack as they have all lost their flagella. An old-starved culture of *V. atlanticus* LGP32 (126 h) significantly immobilized *A. pacificum* ACT03 cells within a few minutes, with lysis occurring immediately (Fig. 3A, C [↗](#)), making it impossible to detect attacks by *V. atlanticus* LGP32 (Fig. 3B [↗](#)). The lysis phase corresponded to initial vesicle formation followed by the bursting of *A. pacificum* ACT03 cells (Movie 5) and was induced by the old-starved culture supernatant of *V. atlanticus* LGP32 (Fig. S1 [↗](#)).

We next tested whether this lytic effect was mediated by thermostable molecule (s) secreted by *Vibrio*. The culture supernatant of starved culture of *V. atlanticus* LGP32 (36 h) filtered through a 10 kDa membrane and then incubated at 100°C for 30 min still possessed its lytic properties, indicating that the algicidal compounds produced by *V. atlanticus* LGP32 are small thermostable molecules unlikely to be lytic enzymes, or lysins able to digest the algae cell. However, these experimental observations clearly show the key role of nutrient limitation in triggering the ‘group-attack’ behaviour and the secretion of lytic compounds of *V. atlanticus* LGP32.

Group attack occurs on *A. pacificum* ACT03 in exponential phase of growth

Here, we wondered whether the live/dead status of algae is important for *V. atlanticus* LGP32-mediated group attacks targeting. To this aim, *A. pacificum* ACT03 in exponential growth phase was first exposed for 30 min to supernatant from 60 hours starved *V. atlanticus* LGP32 Zobell media that induced 25% lysis of *A. pacificum* ACT03 cells and next to the corresponding *V. atlanticus* LGP32 cells. Group attacks were observed on non-degraded *A. pacificum* ACT03 cells, but not on lysed cells. This result is similar to what is observed on the Movie 1 with flash attacks only on immobilized, but not degraded *A. pacificum* ACT03 cells (red), and not on lysed cells (green). In addition, no group attacks occurred on cells from an old *A. pacificum* ACT03 culture (1-month culture). Together, with the very short duration of group attacks (Movie 1), these results indicate that *V. atlanticus* LGP32 group-attacked exponentially growing cells of *A. pacificum* ACT03, but not decomposing algae, suggesting that this behaviour is not just an opportunistic response of heterotrophic bacteria to organic substrates.

Group attack is independent of quorum sensing

Considering the coordinated action of *V. atlanticus* LGP32 in attacks on *A. pacificum* ACT03, we tested whether the group-attack process depended on the key physiological mechanism that regulates many functions in marine microbial cells, quorum sensing (QS) (Lami, 2019 [↗](#); Pappenfort and Bassler, 2016 [↗](#)), a type of cell-cell communication. Although QS is a cell-density-dependent mechanism, our results showed no attack from a 12 h culture of *V. atlanticus* LGP32 up to a concentration of 4.10^6 *Vibrio* mL⁻¹ (Fig 4A [↗](#)). Attacks were only observed with *V. atlanticus* LGP32 from a 60 h culture at low concentration of 5.10^3 *Vibrio* mL⁻¹ to the highest concentration tested of 5.10^5 *Vibrio* mL⁻¹ (Fig 4A [↗](#)), consistent with the hypothesis that the attacks were independent of *Vibrio* density.

The analysis of the expression of genes involved in the known QS pathways in *Vibrio* cell (Fig. S3A [↗](#)), highlighted that only the AI-2 pathway was induced during nutrient stress of *V. atlanticus* LGP32, because only the expression of the AI synthase (*LuxS*) and its receptor (*LuxP*) increased significantly (Fig. 4B [↗](#); ANOVA $p < 0.05$). This was confirmed by a QS bioluminescence assay, which showed a AI-2 molecules (unquantified) in the Zobell culture supernatant of *V. atlanticus* LGP32 after 60 h of culture but not after 12 h of culture and not in the ENSW supernatant of *V. atlanticus* LGP32 after 12 or 60 h of culture. UHPLC-HRMS/MS provided no evidence of detectable HAI-1 and CAI-1 in any experiments. By targeted mutagenesis of key genes involved in QS pathways $\Delta luxM$ (HAI-1 production), $\Delta luxS$ (AI-2 production) and $\Delta luxR$ (high-density QS master regulator) did not lead to any change in the group-attack behaviour of *V. atlanticus* LGP32 (Fig. 4C [↗](#)). Taken together these results showed that group-attack by *V. atlanticus* LGP32 is not coordinated by QS.

Group attack related to the availability of iron

The comparative analysis of the proteome of *V. atlanticus* LGP32 incubated 60 h in artificial seawater (ENSW) versus *V. atlanticus* LGP32 grown 12 h in Zobell nutrient-rich medium revealed 10 proteins modulated by nutrient stress (Fig. S2 [↗](#)). The two most down-regulated proteins correspond to be β -ketoacyl-(acyl-carrier-protein) synthase II (-22-fold in ENSW compared to Zobell), a key regulator of bacterial fatty acid synthesis, and the dihydroorotase (-6.6-fold in ENSW compared to Zobell), an enzyme essential for pyrimidine biosynthesis and thus bacterial proliferation and growth. The low expression of these proteins in ENSW is consistent with *V. atlanticus* LGP32 nutritional starvation. The most up-regulated protein in starved *V. atlanticus* LGP32, with an increase of more than 6-fold, was glucosamine-6-phosphate deaminase, an enzyme involved in bacterial energy metabolism probably necessary for its survival. Among the other up-regulated proteins, one was an iron siderophore-binding protein (Spot 4413, Fig. S2A [↗](#)) corresponding to the vibrioferrin outer membrane receptor PvuB, whose gene is part of the *pvu* operons involved in iron transport (Fig. S3B [↗](#)). Interestingly, the corresponding gene *pvuB* as well as the vibrioferrin membrane receptor gene *pvuA2* (Fig. S3B [↗](#)) were both significantly induced in *Vibrio* under nutrient stress (Fig. 4B [↗](#); ANOVA $p < 0.01$) but not the one involved in the vibrioferrin biosynthesis, *pvsA* (Fig. 4B [↗](#)). Remarkably, among the 10 proteins identified by proteomic analysis only *V. atlanticus* LGP32 mutant lacking *pvuB* failed to attack *A. pacificum* ACT03 (Fig. 4C [↗](#); ANOVA $p < 0.001$). In the absence of the *pvuB* gene, *V. atlanticus* LGP32 was unable to attack in group *A. pacificum* ACT03. In addition, *V. atlanticus* LGP32 cells that had been washed with ENSW to remove their culture supernatant metabolites also failed to attack *A. pacificum* ACT03 (Fig. 4C [↗](#); ANOVA $p < 0.001$), which is congruent with the hypothesis that attacks depend on the *V. atlanticus* LGP32 vibrioferrin transport system. Finally, group-attacks increased significantly, when FeCl₃ was added to the *Vibrio* culture medium (Fig. 4D [↗](#)) but not with H₃BO₄ (Fig. 4D [↗](#)). Taken together, those results are consistent with the hypothesis that attacks are regulated by iron.

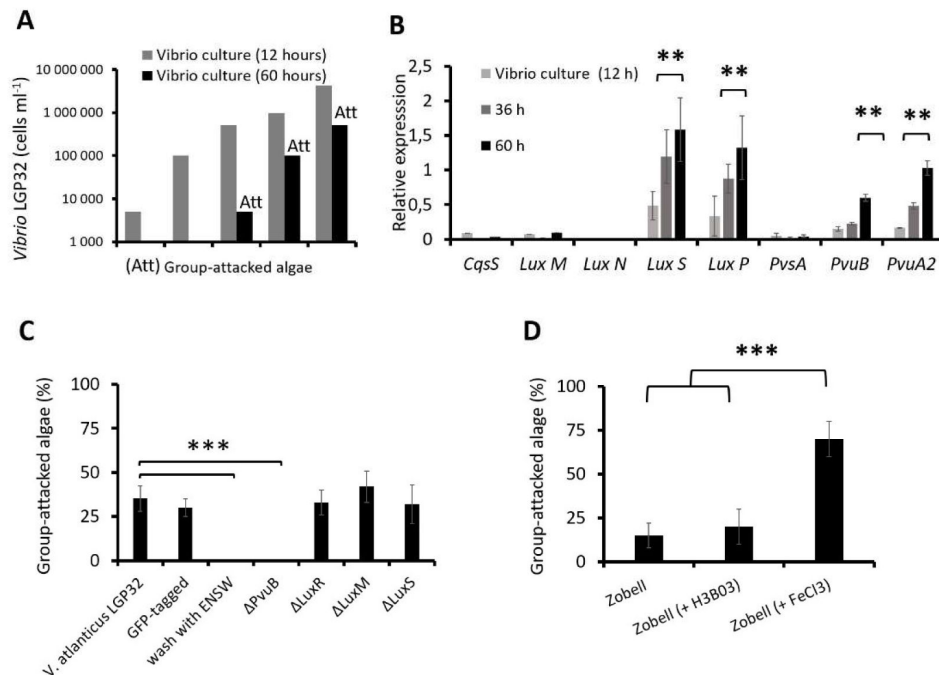


Figure 4.

Role of quorum sensing and the vibrioferrin iron uptake pathway in the interaction process.

(A) Effect of *V. atlanticus* LGP32 cell density on the predation process. Experiments examined the impact of *V. atlanticus* LGP32 cell density on its ability to attack *A. pacificum* ACT03. *A. pacificum* ACT03 cells (2×10^4 cells) were incubated with *V. atlanticus* LGP32 grown for 60 hours in Zobell medium at concentrations ranging from 5×10^3 to 5×10^5 cells mL⁻¹ (represented by black bars). For comparison, *A. pacificum* ACT03 was also incubated with *V. atlanticus* LGP32 grown for only 12 hours in Zobell medium at concentrations ranging from 5×10^3 to 4×10^6 cells mL⁻¹ (represented by grey bars). The term "Att" indicates cells of *A. pacificum* ACT03 that were subjected to group attacks by *V. atlanticus* LGP32. **(B)** CqsS, luxM, luxN, luxS, and luxP quorum sensing and PvsA, PvuB and PvuA2 vibrioferrin pathway genes expression in *V. atlanticus* LGP32 grown for 12, 36 and 60 h in Zobell medium. **(C)** Effect of *V. atlanticus* LGP32 mutants on the group-attacked process. Experiments were conducted by incubating *A. pacificum* ACT03 with *V. atlanticus* LGP32, *V. atlanticus* LGP32 tagged with GFP, *V. atlanticus* LGP32 washed with ENSW or *V. atlanticus* LGP32 mutant ΔPvuB, ΔluxM, ΔluxR and ΔluxS previously grown 60 h in Zobell media (control). The percentage of *A. pacificum* ACT03 group-attacked was determined during the first 30 min of exposure. **(D)** Effect of *V. atlanticus* LGP32 cultures media composition on the group-attacked process. Experiments were conducted by incubating *A. pacificum* ACT03 with *V. atlanticus* LGP32 grown 60 h in Zobell media supplemented with H₃BO₃ or FeCl₃. The results were compared with an exposure to *V. atlanticus* LGP32 grown 60 h in Zobell media. **P < 0.01, ***P < 0.001

Group attack is a *Vibrio* spp. behavior specific to *Alexandrium* spp

To evaluate the dinoflagellates specificity of the group-attack behaviour, a selection of *Vibrio* spp. was co-cultured with a selection of dinoflagellate strains commonly found in the Mediterranean Sea. The results showed that, among the *Vibrio* spp. tested (pathogenic or not) all, under nutrient stress, were able to secrete algicidal compounds, immobilize, ‘group-attack’ and lyse *A. pacificum* ACT03 cells (Table 1) and no tropism linked to their pathogenesis for fish or invertebrates was observed (Table 1). Among the sixteen dinoflagellates species tested, all eight *Alexandrium* spp. (non-toxic and paralytic shellfish toxin (PST) producers) and *Gymnodinium catenatum* (PST producer) were immobilized, ‘group-attacked’ and lysed by *V. atlanticus* LGP32 (Table 2), but no tropism linked to PSTs was revealed (Table 2).

Discussion

Predation is a widespread mode of interaction for survival in the natural world (Finke and Denno, 2004; Sinclair et al., 2003). Among predators, predatory bacteria are found in a wide variety of environments, and like bacteriophages and predatory protists, they have been reported to prey exclusively on other bacteria (Johnke et al., 2014; Perez et al., 2016). Considering predator as a free organism that feeds at the expense of another, this study is the first evidence of the capacity of some *Vibrio* to develop a predatory strategy against an alga. This behaviour differs from parasitism, because the survival of *Vibrio* is not exclusively dependent on the algae.

The strategy developed by *Vibrio* to kill algae may be reminiscent of strategies previously described in the prokaryotes (Johnke et al., 2014). As shown in Movie 1, the interaction between *V. atlanticus* LGP32 and *A. pacificum* ACT03 proceeds in three stages (Fig. 5). The first stage, the ‘immobilization stage’, recalls the strategy used by *Streptomyces* to immobilize its prey (Kumbhar et al., 2014) based on the secretion of algicidal metabolites that disrupt the flagella. The second stage, the ‘attack stage’, resembles the ‘wolf-pack attack’ strategy described for *Myxococcus xanthus* and *Lysobacter* (Martin, 2002; Perez et al., 2014). *V. atlanticus* LGP32 also surrounds *A. pacificum* ACT03 cells at high density for a very short time, but does not invade the algal cell. This phenomenon is comparable to that of bacteria clustering around lysed ciliate cells (Blackburn et al., 1998). The third stage, the ‘killing stage’, is similar to that of epibiotic bacterial predators, which induces the lysis of bacteria from the outside (Rashidan and Bird, 2001). Overall, these observations show that *V. atlanticus* LGP32 is able of wolf-pack hunting behaviour. Results showed that quorum sensing does not control algal predation by *Vibrio*, in contrast to iron acquisition systems, which play a major role in regulating this phenotype, as we demonstrated genetically. Indeed, our quorum sensing mutants exhibited the same predatory phenotype as the wild-type *Vibrio atlanticus* LGP32, whereas mutants impaired in iron uptake completely lost this phenotype. However, quorum sensing and iron acquisition are sometimes interconnected in *Vibrio* (McRose et al., 2018). For instance, in *Vibrio vulnificus*, the production of vulnibactin (a siderophore) is controlled by AI-2 (Kim and Shin, 2011). Whether the production of AI-2 by *V. atlanticus* LGP32, induced during group attacks of *A. pacificum* ACT03, promotes the production of vibrioferriin (Lami, 2019) remains to be investigated.

We further show that the group-attack behaviour of *V. atlanticus* LGP32 depends on iron availability. Iron is an essential element for growth in most organisms, including phytoplankton (Martin and Fitzwater, 1988) and bacteria (Neilands, 1981), and its concentration in seawater is known to be very low with measurements in the open ocean surface at 0.2 nM and in deep waters at 0.6 nM (Millero, 1998).

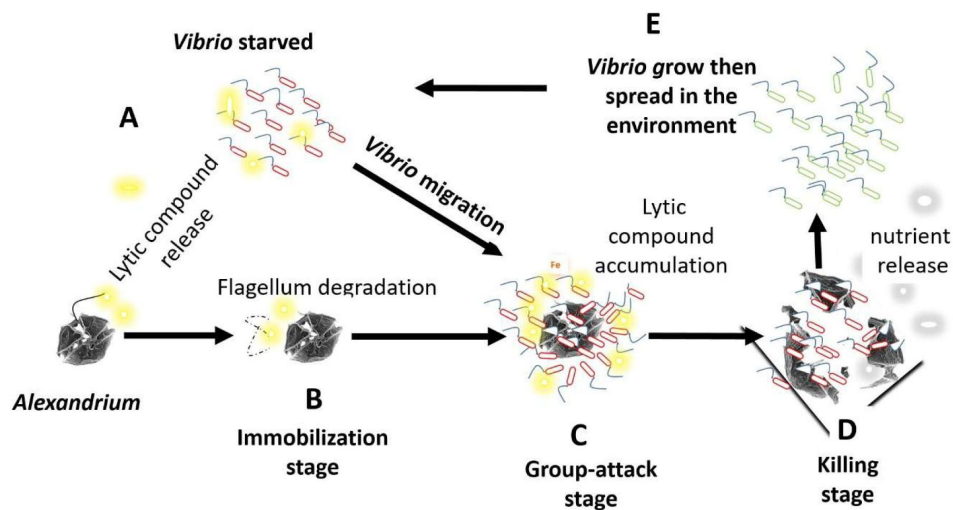


Figure 5.

Schematic representation of a putative strategy developed by *Vibrio* spp. to feed on *Alexandrium* spp. and *G. catenatum* in the environment.

(A) *Vibrio* in the environment when subjected to starvation secrete non-protein lytic compounds. (B) Some of these lytic compounds degrade the flagella, immobilizing the alga (immobilization stage). (C) Then *Vibrio* swims and clusters around its prey coordinating group attacks (group-attack stage). (D) Lytic compounds released by *Vibrio* where able to concentrate around the algae cells, thereby lysing the algae (killing stage). (E) Feeding on the released nutrients, *Vibrio* multiply and then spread in the environment. Yellow clouds: Lytic compound release by *Vibrio*, Grey clouds: Algal nutrients released upon lysis.

Moreover, Its low solubility in seawater limits its availability (Bruland et al., 1994 [↗](#); Wu and Luther, 1994 [↗](#)). To acquire iron, bacteria have developed systems based on the secretion (and subsequent uptake) of iron-chelating siderophores to obtain this element from the environment (Amin et al., 2009a [↗](#)). Therefore, many *Vibrio spp.* produce a siderophore known as vibrioferrin, which is synthesized and secreted by proteins encoded by the *pvsABCDE* gene cluster (Fig. S3 [↗](#)). Boron being known to be a regulator or capable of being transported by vibrioferrin (Romano et al., 2013 [↗](#); Weerasinghe et al., 2013 [↗](#)) we tested its potential involvement in the interaction but no effect was evidenced here. For iron-vibrioferrin uptake, *Vibrio parahaemolyticus* uses a membrane siderophore receptor, called PvuA, which is coupled to an inner membrane ATP-binding cassette (ABC). This ABC transporter system comprised of proteins encoded by the *pvuABCDE* gene cluster (Fig. S3 [↗](#)) is required for transporting the siderophore across the inner membrane (Tanabe et al., 2003 [↗](#)). Siderophores are not only iron carriers but also important regulators of virulence (Miethke and Marahiel, 2007 [↗](#)) and mediators of bacterial interaction with phytoplankton (Amin et al., 2009b [↗](#); Kramer et al., 2020 [↗](#)). We showed here a pivotal role of iron in the interaction between *V. altilanticus* LGP32 and *A. pacificum* ACT03. This mirrors the mutualistic interaction observed between *Gymnodinium catenatum* and *Marinobacter* (Amin et al., 2009b [↗](#)). In fact, in natural settings, the co-occurrence of *Marinobacter* and *G. catenatum* is suggested to depend on a mutually beneficial utilization of iron and carbon resources (Bolch et al., 2011 [↗](#)). As in the present study, iron seems to play a key role in the interaction. Indeed, the labile iron released through the photolysis of ferric chelates with vibrioferrin providing a crucial iron source for phytoplankton, which need substantial amounts of iron to support carbon fixation through photosynthesis (Amin et al., 2009a [↗](#); Yang et al., 2021 [↗](#)). This fixed carbon, in turn, sustains the growth of both the phytoplankton and their bacterial counterparts (Amin et al., 2009b [↗](#); Kramer et al., 2020 [↗](#)). Future study could demonstrate further the role of vibrioferrin in group attack, by adding iron-saturated vibrioferrin to algae-*Vibrio* co-cultures.

In the natural environment, associations between bacteria and algae have already been observed (Lopez-Joven et al., 2018 [↗](#); Miller et al., 2005 [↗](#); Rosales et al., 2022 [↗](#); Xu et al., 2022 [↗](#)). Although we do not have definitive proof of a predator/prey interaction *in situ* due to confounding environmental variables, environmental data acquired in the Thau Lagoon showed a correlation between *Alexandrium* and *Vibrio* occurrence, suggesting that such interaction could occur in nature. Evolution of such a predatory behavior would provide an important ecological advantage to *Vibrio* to obtain nutrients in environment, where *Alexandrium spp.* and *Gymnodinium catenatum* form blooms.

With more than 30 species distributed all over the world (Anderson et al., 2012 [↗](#); Hallegraeff et al., 2021 [↗](#)), *Alexandrium spp.* and *Gymnodinium spp.* considered as invasive species by the Delivering Alien Invasive Species Inventories for Europe (<http://www.europe-aliens.org> [↗](#)), could play an unexpected and important role in maintaining, structuring and regulating *Vibrio* populations in the ecosystem. In turn, *Vibrio* could contribute to the regulation and control of their blooms.

To conclude, this study reveals the capacity of some *Vibrio spp.* to be facultative predators that hunt specific algae. In the current context of climate change favourable to their development monitoring the invasive algae, *Alexandrium spp.* and *Gymnodinium catenatum* should be considered not only for their potent harmful effect on humans and animals, but also as a potential source of nutrients for the expansion of *Vibrio*, particularly pathogenic species (Lemire et al., 2015 [↗](#)).

Data availability

All data on environmental study are included in the article and/or supporting information files. All other study data are available on request from the corresponding author.

Figure S1.

Time and dose-dependent effects of the *V. atlanticus* LGP32 culture supernatant on *A. pacificum* ACT03 motility.

(A) A time dependence experiment was conducted by incubating *A. pacificum* ACT03 for 1, 5 or 20 h with 1/1000 v/v (1 μ L/mL) of culture supernatant from *V. atlanticus* LGP32 previously grown for 60 h in Zobell culture media. (B) A dose dependence experiment was conducted by incubating *A. pacificum* ACT03 for 1 h with 1/1000 to 1/20 v/v (1-50 μ L/mL) of culture supernatant from *V. atlanticus* previously grown for 60 h in Zobell media. The percentage of motile *A. pacificum* ACT03 was determined after 1 hours of exposure. *P < 0.05, ***P < 0.001

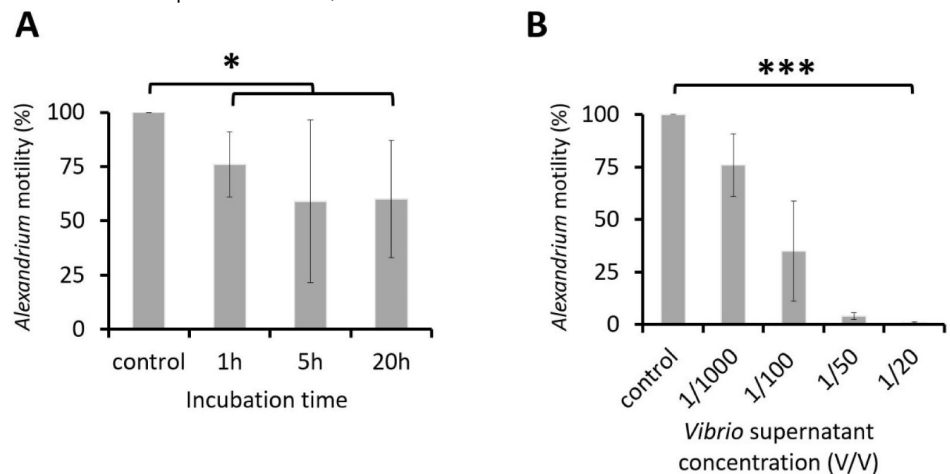
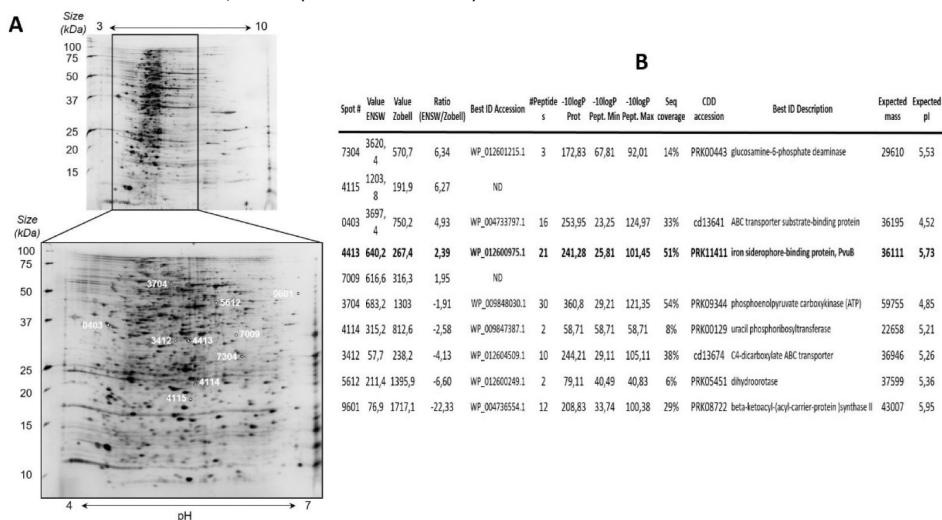


Figure S2.

***Vibrio atlanticus* LGP32 proteome analysis following nutrient stress.**

(A) Example of 2D gel, the numbers in white on the gel 4-7 correspond to the number and position of the protein spots analyzed. (B) Proteins identified by LC-MS/MS as differentially represented in the 2D gel comparative approach following nutrient stress. ND: Not determined; ENSW (artificial seawater).



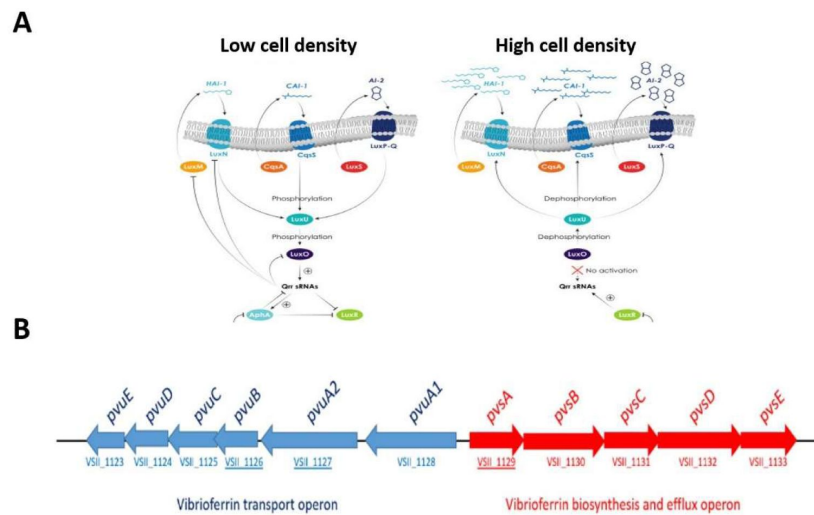


Figure S3.

Quorum sensing and the vibrioferrin iron uptake pathway in *Vibrio*.

(A) Putative quorum sensing (QS) pathways at low and high cell density in *Vibrio* according to Lami et al. (Lami, 2019). **(B)** Genetic organization of the vibrioferrin utilization gene cluster on *V. atlanticus* LGP32 chromosome 2. The Pvu and Pvs operons are involved in the secretion and the transport of ferric vibrioferrin and biosynthesis of vibrioferrin, respectively. Arrows indicate the transcriptional directions of the genes. VSI1126, VSI1137 and VSI1129 corresponding to PvuB, PvuA2 and PvsA genes respectively.

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Additional information

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Additional files

Video 1. Dynamics of *Vibrio atlanticus* LGP32-Alexandrium pacificum ACT03 interaction. GFP-tagged *V. atlanticus* (small green cells); living *A. pacificum* (large red cells); degraded *A. pacificum* (large green cells) filmed under an epifluorescence microscope. [🔗](#)

Video 2. Second-by-second timing of *Vibrio Atlanticus* LGP32 attacking *Alexandrium pacificum* ACT03. GFP-tagged *V. atlanticus* (small green cells); *A. pacificum* living cell (large red cells) filmed under an epifluorescence microscope. [🔗](#)

Video 3. Degradation and disruption of *Alexandrium pacificum* ACT03 flagella. Effect of *Vibrio* supernatant on the first stage of the interaction filmed under a confocal microscope. [🔗](#)

Video 4. Group attacks of *Vibrio atlanticus* LGP32 on target *Alexandrium pacificum* ACT03. *Vibrio*, filmed under a confocal microscope, attacks in groups one immobilized *Alexandrium* cell then moves on to attack — still as a group — another cell without touching the other whole cells, suggesting active communication between *Vibrio* cells. *V. atlanticus* LGP32 (small cells); *A. pacificum* ACT03 (large cells). [🔗](#)

Video 5. Vesicle formation and bursting of an *Alexandrium pacificum* ACT03. Direct effect of *Vibrio* supernatant on *Alexandrium* after 126 h of culture filmed under a confocal microscope [🔗](#)

Data S1. [🔗](#)

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Reviewer #1 (Public review):

Summary:

Rolland and colleagues investigated the interaction between *Vibrio* bacteria and *Alexandrium* algae. The authors found a correlation between the abundance of the two in the Thau Lagoon and observed in the laboratory that *Vibrio* grows to higher numbers in the presence of the algae than in monoculture. Time-lapse imaging of *Alexandrium* in coculture with *Vibrio* enabled the authors to observe *Vibrio* bacteria in proximity to the algae and

subsequent algae death. The authors further determine the mechanism of the interaction between the two and point out similarities between the observed phenotypes and predator-prey behaviours across organisms.

Strengths:

The study combines field work with mechanistic studies in the laboratory and uses a wide array of techniques ranging from co-cultivation experiments to genetic engineering, microscopy and proteomics. Further, the authors test multiple *Vibrio* and *Alexandria* species and claim a wide spread of the observed phenotypes.

Weaknesses:

In my view, the presentation of the data is in some cases not ideal. The phrasing of some conclusions (e.g., group-attacks and wolf-pack-hunting by the bacteria) is in my opinion too strong based on the herein provided data.

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Reviewer #2 (Public review):

Goal summary:

The authors sought to (i) demonstrate correlations between the dynamics of the dinoflagellate *Alexandrium pacificum* and the bacterium *Vibrio atlanticus* in natural populations, ii) demonstrate the occurrence of predation in laboratory experiments, iii) claim coordinated action by the predators in the predation process, iv) demonstrate that predation is induced by predator starvation, and v) test for effects of quorum sensing and iron-uptake genes on the predation process.

Strengths include:

- (1) Data indicating correlated dynamics in a natural environment that increase the motivation for the study of in vitro interactions.
- (2) Experimental design allowing clear inference of predation based on population counts of both prey and predators in addition to microscopy-based evidence.
- (3) Supplementation of population-level data with molecular approaches to test hypotheses regarding possible involvement of quorum sensing and iron uptake in predation.

Weaknesses include:

- (1) A lack of early, clear definitions for several important terms used in the paper, including 'predation', 'coordination' and 'coordinated action', 'group attack', and 'wolf-pack hunting', along with a corresponding lack of criteria for what evidence would warrant use of some of these labels. (For example, does mere simultaneity of attacks of an *A. pacificum* cell by many *V. atlanticus* cells constitute "coordination"? Or, as it seems to us, does coordination require some form of signalling between predator cells?)
- (2) Absence of controls for cell density in the test for starvation effects on predatory behavior; unclear how the length of incubation affects the density of *V. atlanticus* cells.
- (3) Lack of clarity in some of the methodological descriptions

Appraisal:

The authors convincingly achieve their aim of demonstrating that *V. atlanticus* can prey on *A. pacificum*, provide strongly suggestive evidence that such predation is induced by starvation, and clearly demonstrate that both iron availability and, correspondingly, the presence of genes involved in iron uptake, strongly influence the efficacy of predation. However, the evidence for starvation-induction of predation can be strengthened with cell-density controls; evidence for a social component to predation - positive interactions between attacking predators - is lacking.

Discussion of impact:

This paper will interest those interested in how microbial behaviour responds to environmental fluctuations, in particular predatory behaviour, but will do so more strongly if the evidence of starvation-induction of predation is strengthened. It will also interest those investigating bacteria-algae interactions and potential ecological controls of algal blooms. It has the potential to interest researchers of microbial cooperation, should the authors be able to provide any evidence of coordination between predator cells.

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